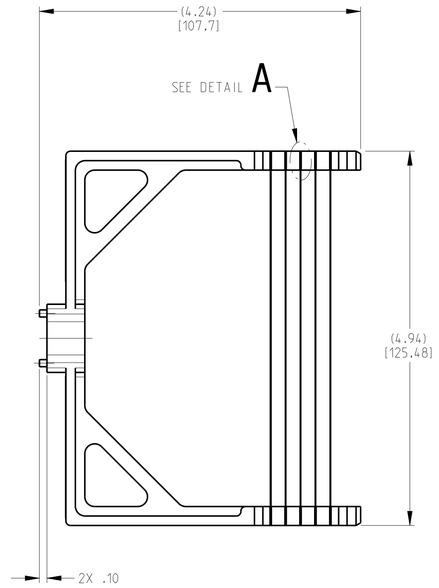
DRAWING LIST

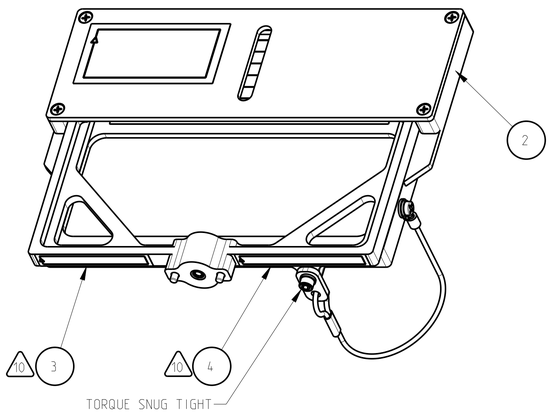
ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	S1101MFD1538	TFP ARRAY BRACKET
2	1	S1101MFA1539	TFP ARRAY COVER ASSEMBLY
3	1	S1101MFT1558-01	ID LABEL
4	1	S1101MFT1558-XX	PART AND SERIAL NUMBER LABEL

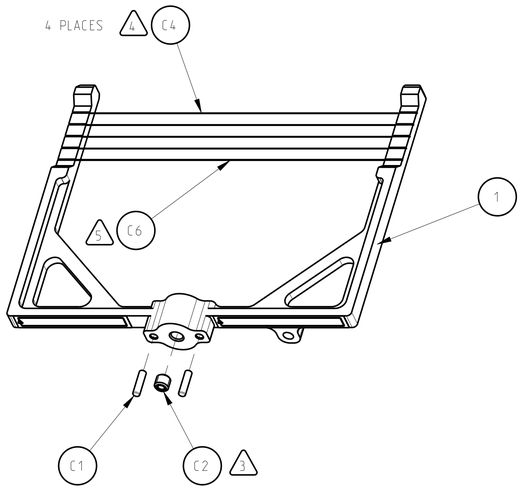
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	2	MS9390-090	PIN, STRAIGHT, HEADLESS, DOWEL, CRES STANDARD, .09375 X .375LG
C2	1	MS51830CA102	MINIATURE INSERT FOR .112-40 UNJC-3A IN. FASTENER
C3	A/R	1838 B/A GREEN	3M SCOTCH WELD EPOXY ADHESIVE
C4	A/R	SCS-6	140 MICROMETER SILICON CARBIDE FIBER, SPECIALTY MATERIALS
C5	A/R	49550	LOCTITE 495 SUPER BONDER INSTANT ADHESIVE
C6	A/R	PT005130	.20mm DIA. PLATINUM WIRE, GOODFELLOW

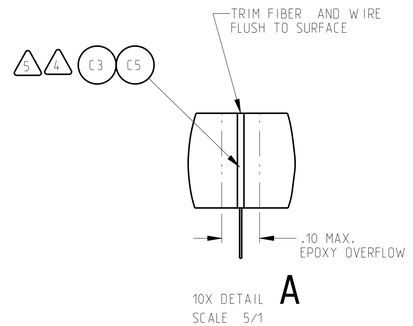


NOTE: TFP ARRAY COVER ASSEMBLY NOT SHOWN FOR CLARITY





NOTE: TFP ARRAY COVER ASSEMBLY NOT SHOWN FOR CLARITY



10X DETAIL A
SCALE 5/1

NOTES:

1. MATERIAL AND PROCESS CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ASSEMBLE PER APPROVED PROCESS PLAN FOR INITIAL BUILD.
3. INSTALL THREADED INSERTS PER MS51835. VERIFY INSERT IS BELOW SURFACE BEFORE SETTING KEYS.
4. FIBER ATTACHMENT PROCEDURE (4X):
   - A. REMOVE A SINGLE FIBER FROM ITEM C4. LAY FIBER IN POSTION.
   - B. APPLY ADHESIVE ITEM C5 INTO FIRST FIBER GROOVE & LET CURE.
   - C. APPLY 7.8±.1 GRAM TENSILE LOAD ON OPPOSITE END OF FIBER AND SET INTO GROOVE. APPLY ADHESIVE ITEM C5.
   - D. LET CURE FOR 5 MINUTES MINIMUM. REMOVE LOAD AND TRIM FIBER TO LENGTH.
   - E. APPLY EPOXY ITEM C3 TO COVER GROOVE AND FIBER CONTACT AREAS.
5. WIRE ATTACHMENT PROCEDURE (1X):
   - A. TRIM A PIECE OF WIRE (ITEM C6) TO 9" AND LAY INTO POSITION.
   - B. APPLY ADHESIVE ITEM C5 INTO LAST GROOVE & LET CURE.
   - C. APPLY 35-40 GRAM TENSILE LOAD ON OPPOSITE END OF WIRE AND SET INTO GROOVE. APPLY ADHESIVE ITEM C5.
   - D. LET CURE FOR 5 MINUTES MINIMUM. REMOVE LOAD AND TRIM WIRE TO LENGTH.
   - E. APPLY EPOXY ITEM C3 TO COVER GROOVE AND WIRE CONTACT AREAS.
6. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3 (GOLD COLOR).
7. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
8. HANDLE WITH CLEAN, LINT-FREE GLOVES.
9. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, TRANSPARENT POLYETHYLENE BAG FOR SMALL ITEMS OR 4 MIL THICK MINIMUM POLYETHYLENE SHEET AND SEAL. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW, BY TAGGING PER SAE-AS478-35D.
10. APPLY LABEL TO CLEAN, DRY SURFACE IN LOCATION SHOWN.

REV	DESCRIPTION	APPROVED
B	INCORPORATED ACME-ECO-514 AND ACME-ECO-587.	AK 11/07/16 LF 11/07/16 AD 11/07/16
A	INCORPORATED ACME-ECO-428	AD 03/30/15 JO 03/30/15 GF 03/26/15

REVISION HISTORY

S1101MFA0100
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± .03
   .XX ± .01
   .XXX ± .005
5. ANGULAR TOLERANCE ± .5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED:
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	K.RHODES 11/10/14	QUALITY	D.SEELEY 11/17/14
REVIEWER	G.FEDEVICH 11/12/14	MATERIALS	M.WILKINSON 11/10/14
MECHANICAL	A.DRAKE 11/13/14	PROJ LEAD	B.BOROWSKI 11/17/14
ELECTRICAL	T.GOBELI 11/10/14		
STRUCTURAL	C.WEAVER 11/14/14		
SAFETY	R.BORRELLI 11/12/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

ZIN TECHNOLOGIES
8745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
140 MICRON TFP CALIBRATION ARRAY ASSEMBLY

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA1558	B

SCALE 1/1 & NOTED | INITIAL REL DATE 11/17/14 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL: S1101MFA1558

PRO/E VERSION: Wildfire 5.0

REV B | SH 1 | DWG NO S1101MFA1558

ZIN_D_2012.frm